



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
SEMESTER I, SESSION 2024/2025

BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECV 2113 – HUMAN COMPUTER INTERACTION

EV CHARGING STATIONS TRACKER

TASK 3

GROUP: NOBODY

GROUP MEMBERS	MATRIC NO
TAN ZHAO HONG	A23CS0188
CHIN PEI WEN	A23CS0065
KOO XUAN	A23CS0300
LING YU QIAN	A23CS0301
NAZATUL NADHIRAH BINTI SABTU	A23CS0144

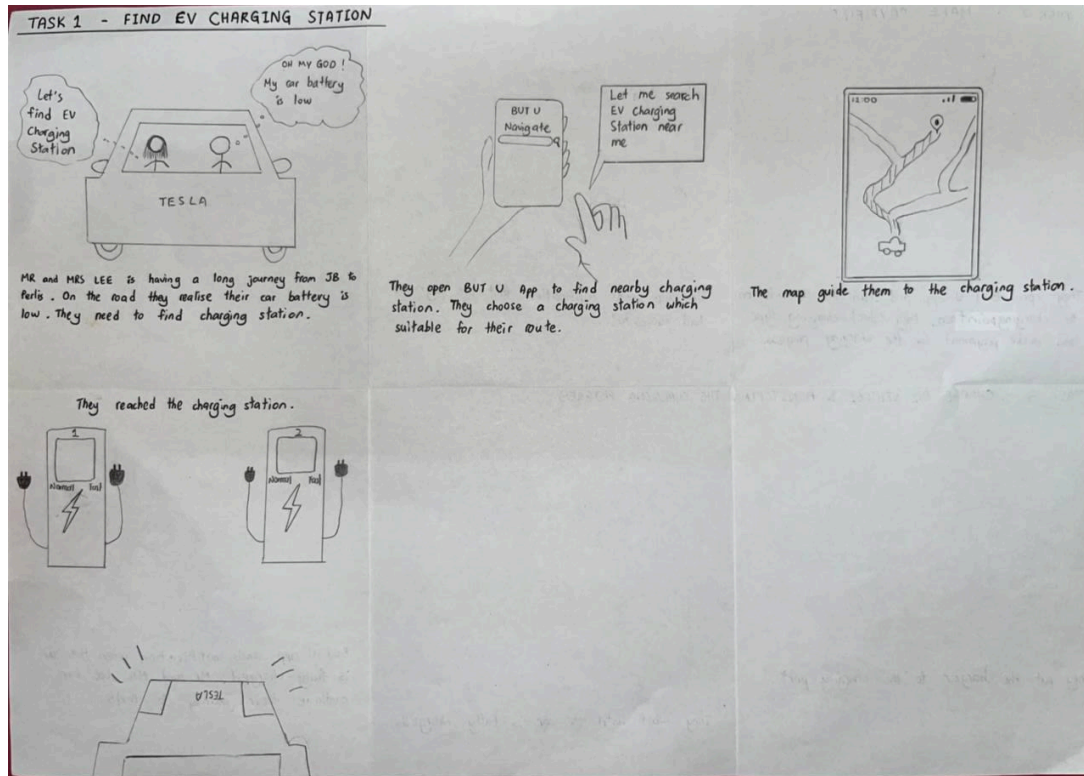
Lecturer's name : Dr. Nur Zuraifah Syazrah binti Othman

Title: Conceptual & Physical Design

Storyboard

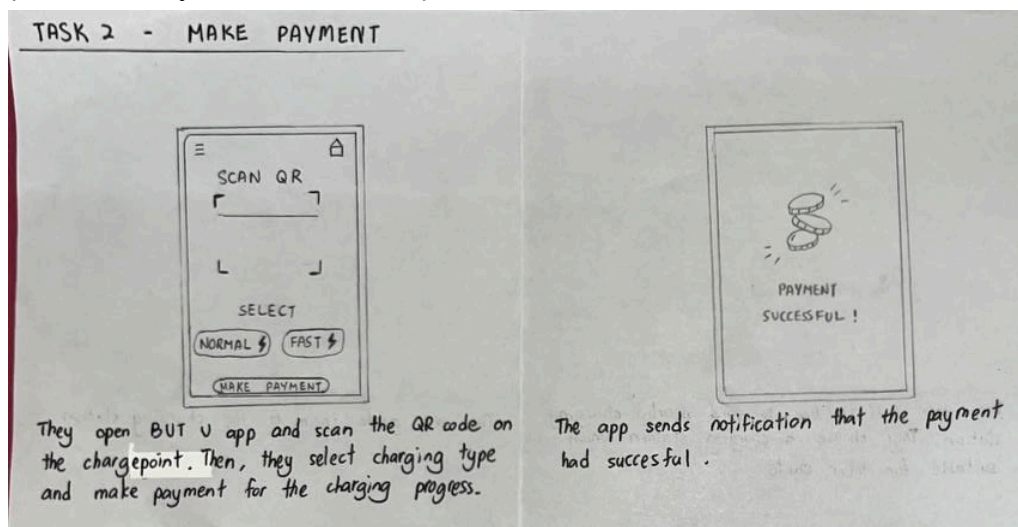
Task 1-Choosing / Finding Charging Station

{Scan of storyboard for Task 1}



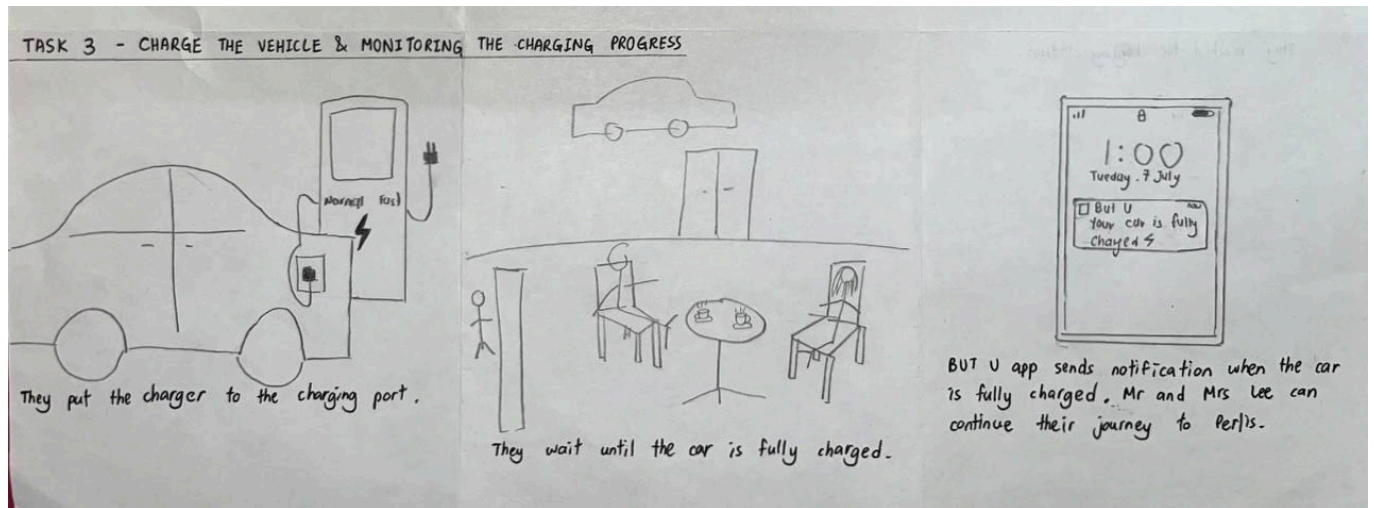
Task 2-Make Payment

{Scan of storyboard for Task 2}



Task 3-Charge the vehicle and monitor charging progress

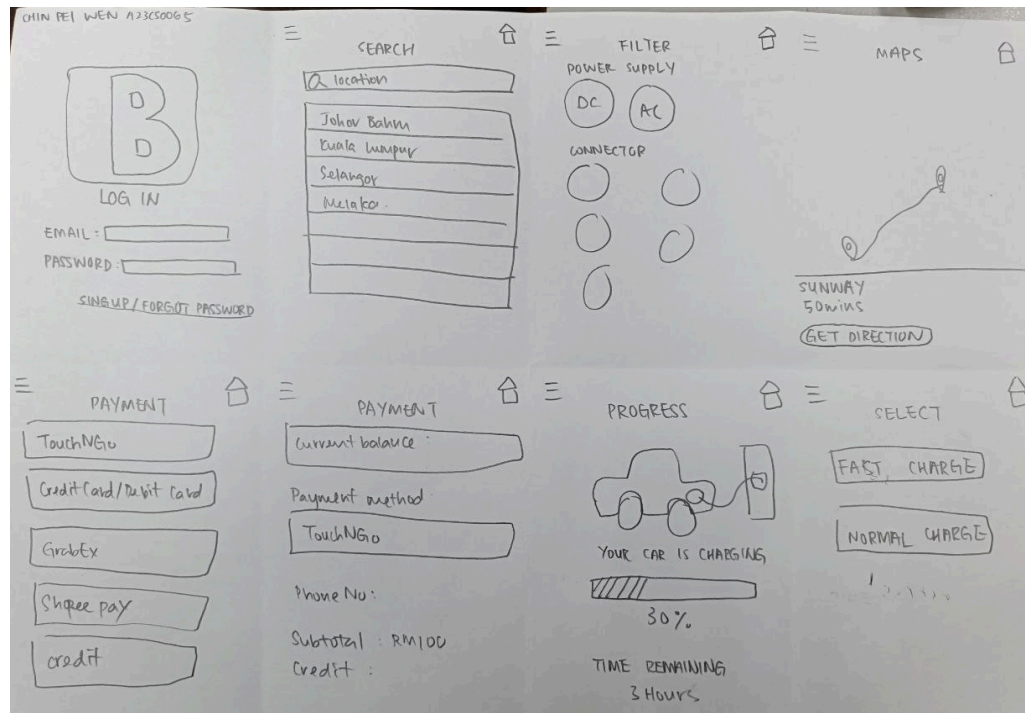
{Scan of storyboard for Task 3}



Alternative Design

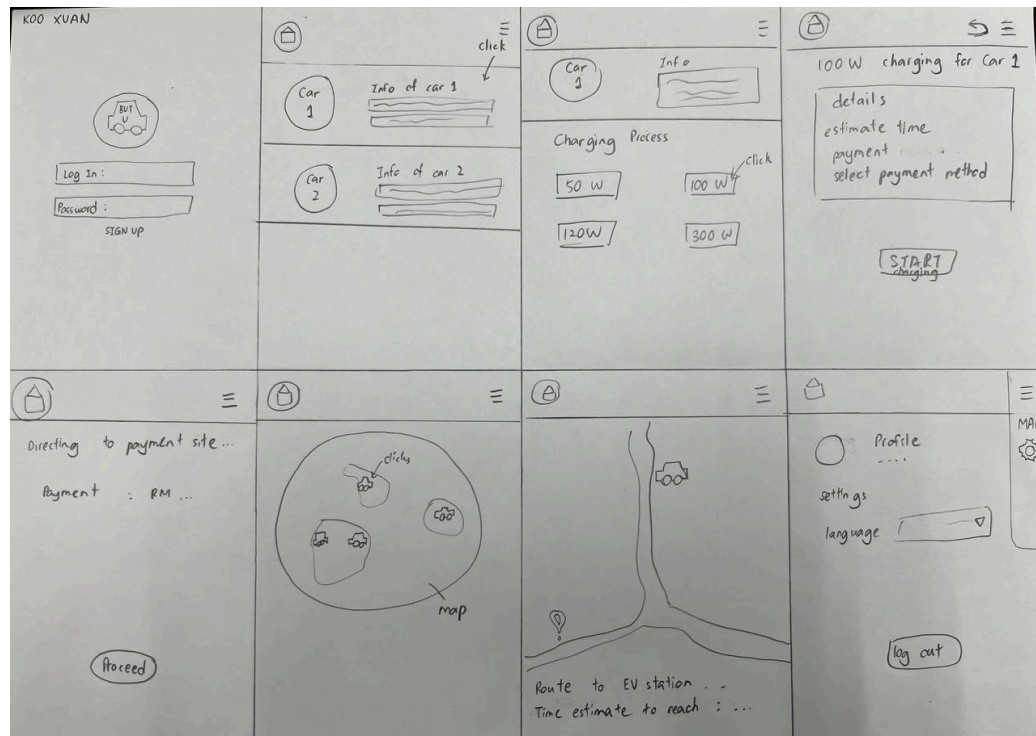
Alternative Design 1 – CHIN PEI WEN

{Scan of Crazy 8 from Member 1}

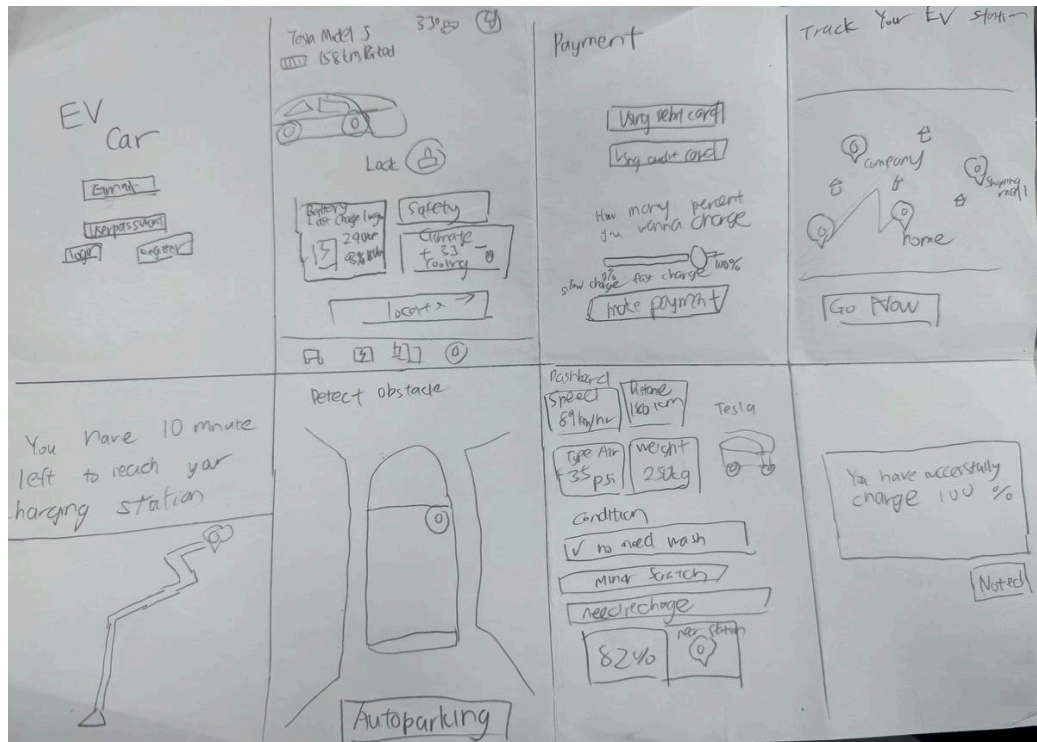


Alternative Design 2 – KOO XUAN

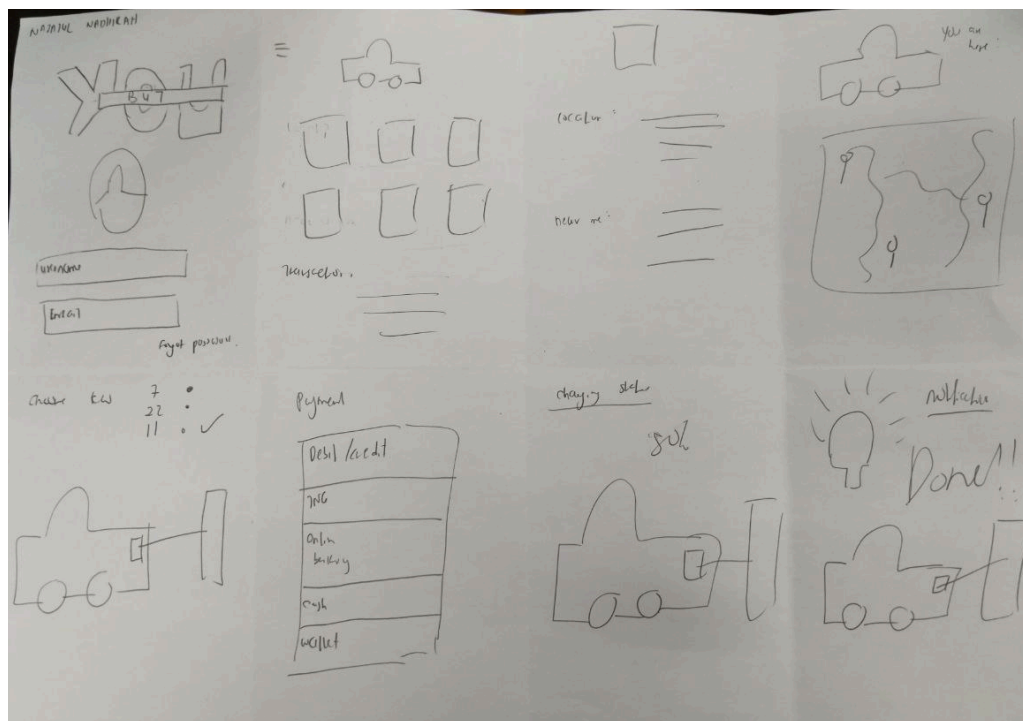
{Scan of Crazy 8 from Member 2}



Alternative Design 3 – LING YU QIAN {Scan of Crazy 8 from Member 3}



Alternative Design 4 – NAZATUL NADHIRAH BINTI SABTU {Scan of Crazy 8 from Member 4}



Alternative Design 5 – TAN ZHAO HONG

{Scan of Crazy 8 from Member 5}

TAN ZHAO HONG ASSOCIATES

But V 

Sign In

Email: _____

Password: _____

Forgot your password?

Nearby charging station

Find

10

20

30

40

50

60

70

80

90

100

Payment method

Credit/Debit card

Online banking

Filter

Charger type

DC

AC

Charger power

7kW

11kW

Rate

Navigation to Google map to get route

Direction

Account setting

Edit

Name

Email

Mobile Number

Current balance

Add Credit

Current balance

Select amount to add

10

20

30

40

50

60

70

80

90

100

Payment method

Credit/Debit card

Online banking

Charging progress

Battery level

70%

Desired battery level

Estimated time

Help

Need help now?

Call us

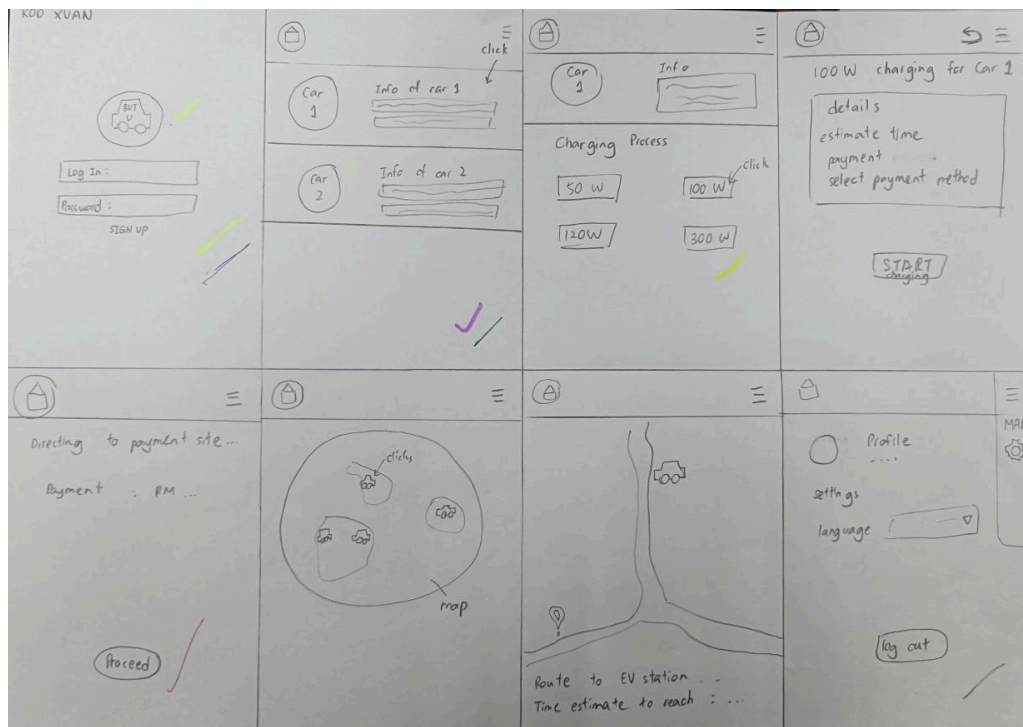
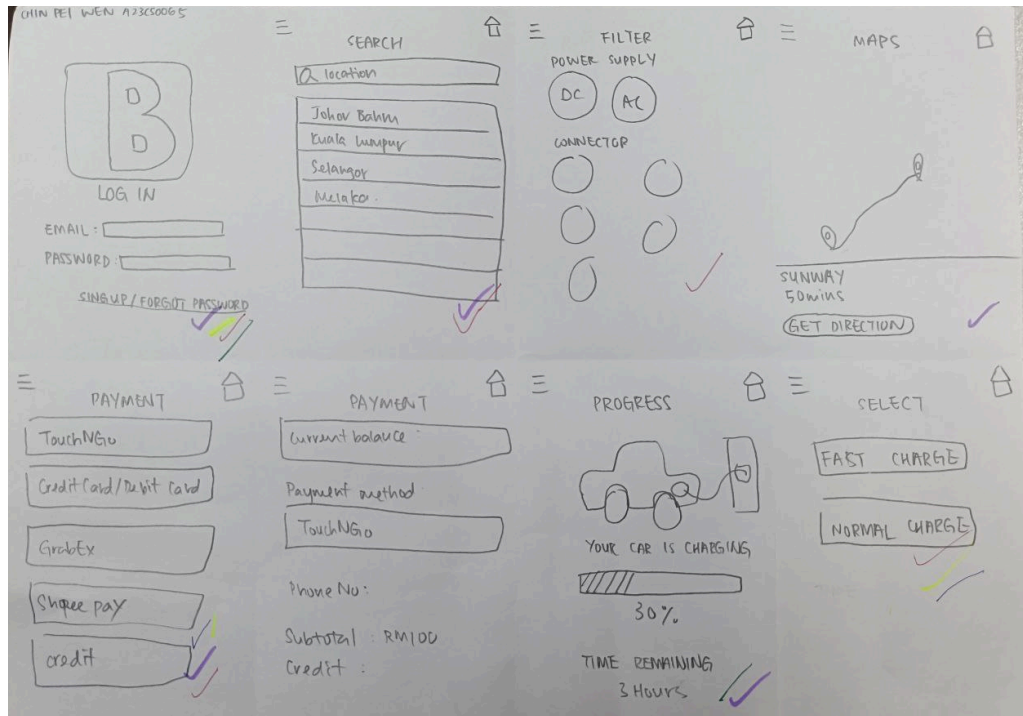
Tutorial

Read FAQ

Formal as as feedback@gmail.com

{Scan of voted design elements/layouts}

CHIN PEI WEN, KOO XUAN, LING YU QIAN, NAZATUL, TAN ZHAO HONG



EV Car

Email

Accession

Lock

Unlock

Test Model S 330.80

158 km/h

Lock

Unlock

Location

Payment

Using debit card

Using credit card

How many percent you wanna charge

show charge for charge

make payment

Track Your EV status

company

home

Go Now

You have 10 minute left to reach your charging station

Detect obstacle

Autoparking

Portland

Speed 81 km/hr

Home 10 km

Testa

Gre Air 35 psi

Weight 28 kg

Condition

✓ no need wash

Minor scratch

Need recharge

82% per cent

You have successfully charge 100 %

Noted

YOU

YOU

YOU

YOU

YOU

YOU

YOU

YOU

YOU

YOU

YOU

YOU

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YOU

YOU

YOU

YOU

IAN ZHANG HONG ASSOCIATES

But U 

Sign In

Email: _____

Password: _____

Forgot your password?

Nearby charging station

A Find

☐ AC
☐ DC
☐ Power
☐ Port
☐ Range

Filter

Charger type

☐ DC ☐ AC

Charger power

☐ 7kW ☐ 11kW

Rate

(Navigate to Google map to get rate)

Direction

Account setting

Edit

Name _____

Email _____

Mobile Number _____

Current balance _____

Add Credit

Current balance _____

Select amount to add:

☐ 10 ☐ 50 ☐ 80

☐ 150 ☐ 1000 ☐ 2000

Payment method:

☐ Credit/Debit Card

☐ Online Banking

Charging progress



Battery level: ☐ 70%

Desired battery level: _____

Estimated time: _____

☐ Fast ☐ Normal

Help

Need help now?

☐ Call us

or:

☐ Tutorial

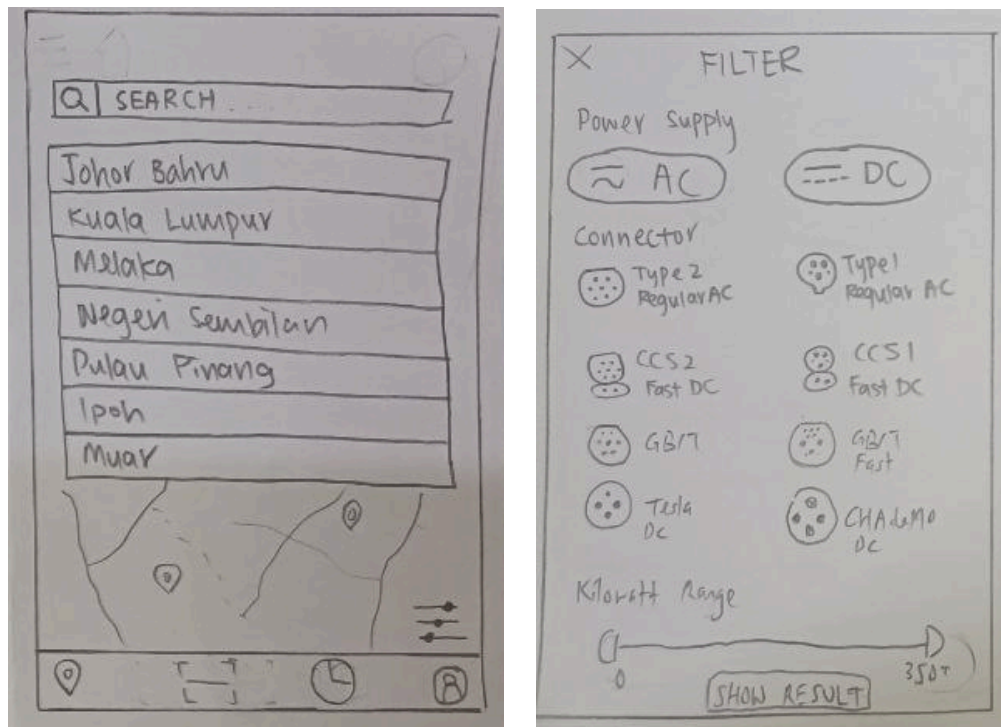
☐ Read FAQ

☐ Email us as bahy@gmail.com

{Wireframes for Task 1, 2 and 3 - should consist of one or more sketched interface controls that the user can manipulate and interact with, and roughly show the flow of the task}

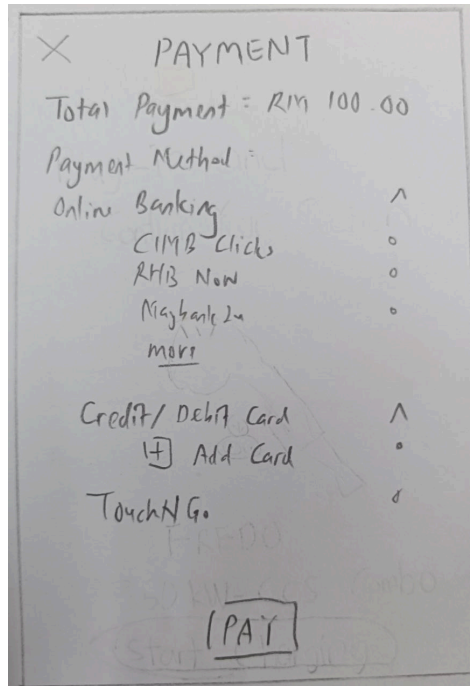
Wireframe for Task 1 - Find charging station

Search the location that the user wants to go on the searching bar, then it will show all available charging stations on the map. After that, the user can choose the charging station and get directions to reach there. Users can filter the result by choosing their preferred charging requirement.



Wireframe for Task 2 - Making payment

After the user confirms charging, it will show the total payment needed for the charging process. Then, it will show a lot of payment methods such as online banking, credit/debit card and TouchNGo. Users can choose any payment method to make payment.



Wireframe for Task 3 - Track charging progress

During the charging process, users can track the battery level of the vehicle. Users also can view the remaining time for complete charging. After the charging process is complete, a notification is sent to the user.

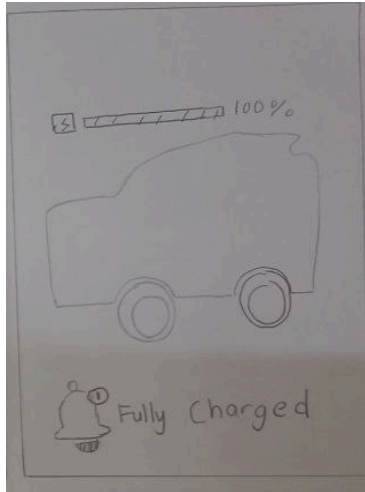
X

⚡  88%



range: 201 km 352 kW

⚡ Time remaining: 1h01min



{Reason or Justification of the Design}

Similarity

Each feature icon is similar in size and color. Each functional button also maintains the same size, ensuring consistency in styling of our contents.

Figure-Ground

All feature buttons are designed to look like press-able buttons by making all the buttons stand out against the background. This can increase readability and usability of the features.

Good Continuation

Navigation flows in a logical, step-by-step manner. For example, This smooth, directional flow makes it intuitive for users to progress through tasks without confusion.

Closure

Icons such as car outline for charging status, QR code placeholders are still recognizable. Even without complete details, the human brain naturally fills in gaps, enhancing visual clarity without cluttering the design.

Simplicity

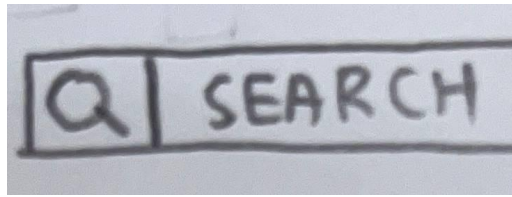
The design adopts a minimalistic layout with clean lines, limited text, and consistent icons. Reducing visual noise makes it easier for users to quickly identify and interact with essential buttons.

Experience

The design accommodates a clear user flow from searching for charging stations to payment. Icons for filters, maps for navigation, and steps for user authentication simplify user experience. These user-friendly design elements enhance usability and meet the expectations of a tech-savvy audience.

{Description and justification of metaphors}

1. Magnifying Glass Icon for Search Function



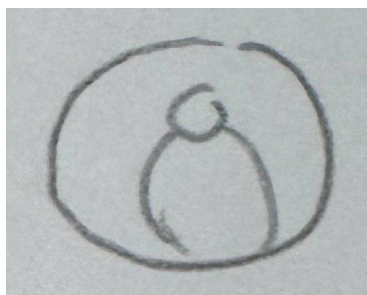
The magnifying glass icon on the search screen symbolizes the search action, just as a magnifying glass helps people look closely to find details in the real world. This metaphor is widely used in digital interfaces, and users associate it with finding information or specific locations. In your design, it simplifies locating charging stations or places of interest.

Clock for History



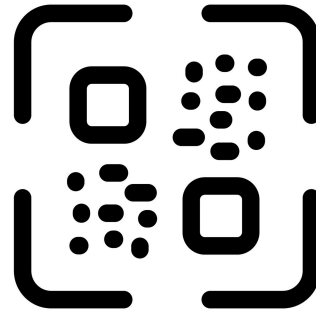
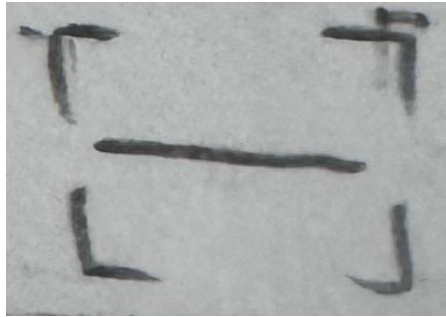
The clock icon symbolizes time tracking and transaction history, similar to viewing a timeline in real life. This metaphor reflects real-world receipts and time records, allowing users to review past sessions.

Profile Avatar for User Identity



The avatar icon on the profile page represents user identity, much like a personal photo or ID card in the real world. This metaphor enhances personalization, helping users connect with their profiles and reinforcing a sense of account ownership.

QR Code for Seamless Connection



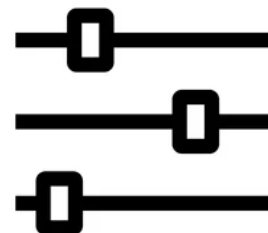
The QR code metaphor represents a digital key or shortcut to perform actions such as scanning for payments or initiating charging. It mimics real-world barcode scanning, where users quickly access information. This metaphor eliminates manual inputs and enhances convenience for users.

Location Pin and Map for Navigation



The location pin icon symbolizes the real-world action of marking or pinpointing a location. This metaphor reflects physical maps or navigation apps, where a pin guides users to their desired location. In my design, it helps users find charging stations and navigate easily to them.

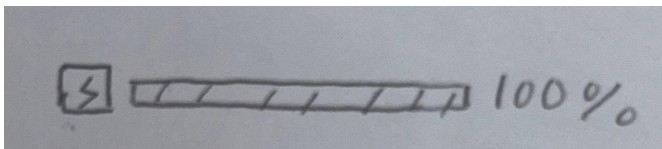
Adjustment Slider Icon for Filter Function



The adjustment slider icon symbolizes the process of refining and customizing options, similar to adjusting controls on physical devices like audio equalizers or mechanical dials. This metaphor

represents precision and flexibility, allowing users to fine-tune their preferences. The adjustment slider icon helps users filter charging options by power supply (AC/DC), connector type (e.g., CCS2, Tesla), and range, providing a streamlined and intuitive experience for locating the most suitable charging station.

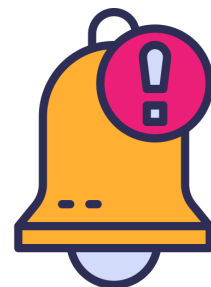
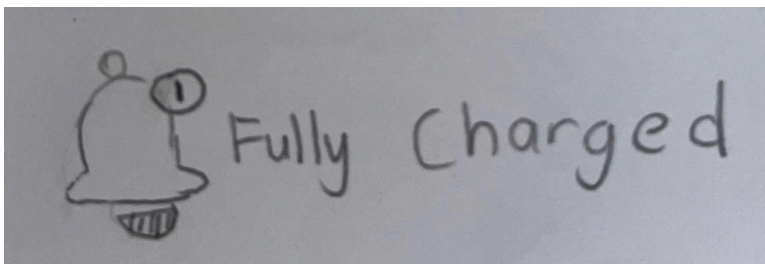
2. Fuel Tank for battery



The charging status of the EV is represented using a metaphorical fuel tank icon, where the battery's percentage is depicted as a filling bar. The remaining charging time is shown as a countdown timer, similar to a fuel gauge in traditional vehicles.

Justification: The metaphor of a fuel tank resonates with users transitioning from traditional vehicles to EVs. It offers a clear, intuitive way to understand battery levels and charging progress without requiring extensive learning.

3. Notification Bell for Alerts (Attention Metaphor)



Description: A notification bell metaphor is used to deliver updates and alerts about the charging process, such as when charging is complete, interrupted, or if the vehicle is disconnected.

Justification: This metaphor draws on the widely recognized use of notification bells in digital interfaces. It ensures that users remain informed about critical updates without actively monitoring the app, thus enhancing convenience.

